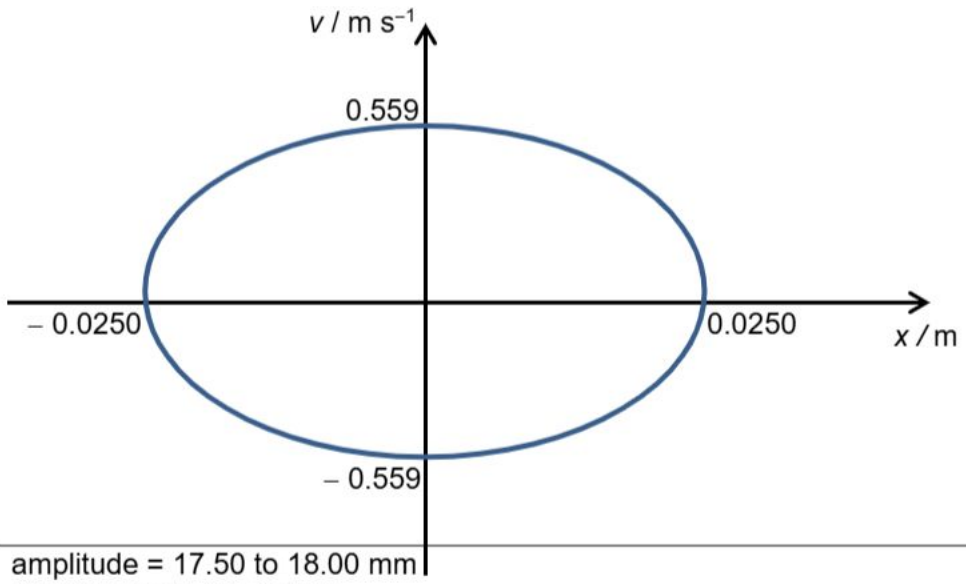
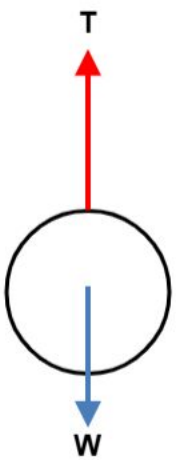
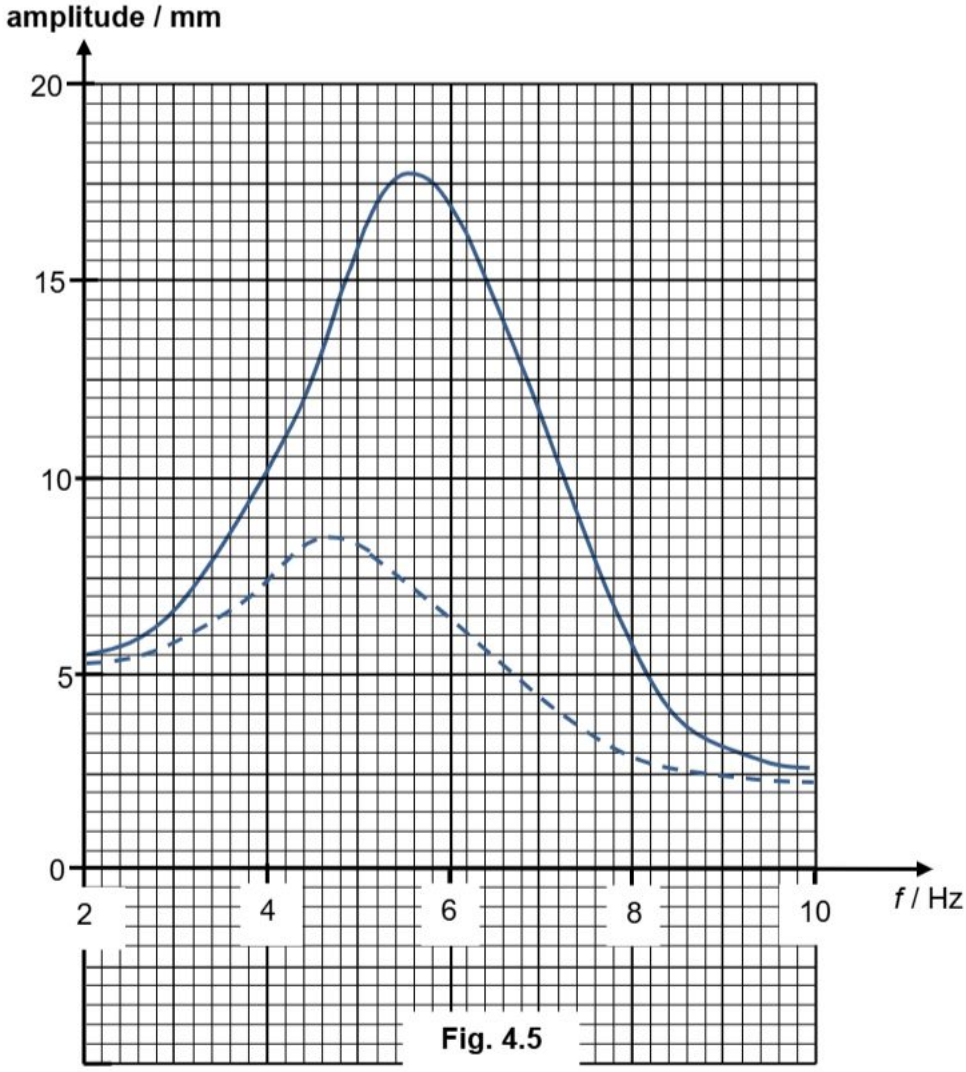
$KE_B \quad M_1$

4(a)(i)	The graph is a **straight** line that passes through the **origin**. This indicates that the acceleration is **directly proportional** to the displacement, which is a characteristic of simple harmonic motion. The gradient of the graph is **negative**. This indicate that the direction of the acceleration and the displacement are in **opposing direction** **or** the acceleration is directed to a **fixed equilibrium position**.	B1 B1

(a)(ii)	From the graph, x_o = 2.50 cm = 0.025 m a_o = 12.50 m s^{-2} $a_o = \omega^2 x_o$ $\Rightarrow \omega^2$ = 12.50 / 0.0250 = 500 s^{-2} $v_o = x_o\omega$ = 0.559 m s^{-1}  **Fig. 4.3**	C1 (value of x_o) A1 (value of v_o) A1 (shape of graph)
(b)(i)	amplitude = 17.50 to 18.00 mm frequency = 5.4 to 5.6 Hz - both answers correct - both presented to correct d.p.	A1
(b)(ii)1.	W: Weight of ball T: Tension acting on the ball by the string  Note: T is upwards, W is downwards, and T must be larger than W	B1

(b)(ii)2.	At the maximum tension, the net force on the ball must be maximum. By Newton's second law of motion, $F_{net} = T_{max} - W$ $\therefore T_{max} = F_{net} + W$ $= ma_o + mg$ $= m(w^2x_o) + mg$ $= m(2\pi f)^2x_o + mg$ $= (0.150)[2\pi(5.6)]^2(0.0177) + (0.150)(9.81)$ $= 4.76$ N	C1 M1 A1
(b)(iii)	 **Fig. 4.5**	B1 (similar shape: peak flatter) B1 (amp lower for all freq AND peak freq lower)
5(a)	A progressive wave is a wave in which <u>energy is carried from one point to</u>	